

Executive Summary

The **Jere Model Academy School Management System (SMS)** is a comprehensive, modern, web-based digital portal engineered to streamline school administration, automate academic assessment workflows, manage fee collections, and provide seamless communication channels for Administrators, Teachers, Students, and Parents. Designed to support Nursery, Primary, Junior Secondary (JSS), and Senior Secondary (SSS) tiers, the system offers an intuitive user experience paired with robust security, digital result protection, and rich analytics.

System Architecture & Role-Based Portals

User Role	Primary Responsibilities & Access Level	Key Portal Features
System Administrator	Full operational governance, school configuration, user account management, and financial control.	• Student & Staff Directory • Fee Structure & Payment Tracking • Result PIN Generation • Broad-Sheet & Report Card Audits • Session & Term Management • System Settings & Branding
Teacher / Form Master	Class-level academic management, score entry, attendance tracking, and behavioral evaluation.	• CA & Exam Score Entry • Class Attendance Marking • Form Teacher Remarks • Affective & Psychomotor Evaluation • Class Broadsheets View • Scheme of Work Publishing
Student / Parent	Personal academic portal for checking termly results, tracking payments, and accessing curriculum.	• Secure Result Checking (PIN) • Downloadable Report Cards • School Fee Statements & Receipts • Interactive Scheme of Work • School Rules & Digital Undertakings

Key System Functional Modules

1. Student Information System (SIS) & Admission Management

- **Comprehensive Profiles:** Stores complete student records including Admission Numbers, Date of Birth, Gender, State of Origin, LGA, Residential Address, and Passport Photographs.
- **Parent & Guardian Integration:** Keeps emergency contact information, parent phone numbers, and home address details.
- **Special Needs & Health Tracking:** Dedicated fields for physical handicaps or special care requirements.
- **Academic Class Assignment:** Flexible placement across Nursery, Primary, JSS, and SSS classes and arms.
- **Student Promotion & Archiving:** Smooth session-to-session class promotion and record status management (Active, Suspended, Archived).

2. Academic Assessment & Examination Engine

- **Flexible CA & Exam Weightings:** Fully customizable score structures (CA1, CA2, CA3, CA4, and Termly Exams) tailored per school specification.
- **Automated Score Calculation:** Real-time aggregation of continuous assessments and exams with automatic total calculations and grade assignment (A, B, C, D, E, F).
- **Affective & Psychomotor Evaluation:** 5-point rating scales for behavioral traits (Honesty, Punctuality, Neatness, Leadership) and practical skills (Sports, Crafts, Handwriting).
- **Form Teacher & Principal Remarks:** Integrated comment engine providing personalized qualitative feedback on every student's report card.
- **Class Broadsheets:** Complete master broadsheets displaying subject performance side-by-side for an entire class for academic review.

3. Official Report Cards & Bulk Printing

- **Digital Report Cards:** Clean, printable report card documents containing school branding, logo, principal signature, position in class, class average, attendance summary, and term resumption dates.
- **One-Click Bulk Result Printer:** Allows administrators to generate and print report cards for an entire class simultaneously, saving hours of manual administrative work.
- **Digital Signature Integration:** Electronic signature capture and rendering for Teachers and Principals to authenticate issued report cards.

4. Scratch Card Style Result PIN Security System

- **Anti-Tamper Result Checking:** Unique 12-character alphanumeric PIN codes generated specifically for each student and term.
- **Usage Limit Controls:** Built-in limit enforcement (e.g. maximum 5 checks per PIN) preventing unauthorized distribution or misuse.
- **Financial Control:** Result PINs can be issued upon fee clearance, ensuring parents fulfill financial obligations before retrieving report cards.

5. School Fee Management & Digital Receipting

- **Tiered Fee Structures:** Custom fee items defined by school tier (Nursery, Primary, JSS, SSS) covering Tuition, Books, Uniforms, and Development levies.
- **Student Financial Ledgers:** Automatic tracking of Amount Due, Total Paid, and Outstanding Balance per student.
- **Multiple Payment Log Types:** Supports Cash, Bank Transfer, and POS transactions logged by bursars/administrators.
- **Official Digital Receipts:** Instant generation of unique receipt numbers and printable payment proof for parents.

6. Attendance Tracking Module

- **Daily Attendance Registers:** Teachers can mark student status (Present, Absent, Late) in seconds.
- **Report Card Integration:** Total days present and total days school opened automatically calculate and reflect on final report cards.

7. Scheme of Work & Academic Curriculum Delivery

- **Weekly Curriculum Publishing:** Teachers post weekly topics, subtitles, and learning objectives for each subject and class.
- **Student Portal Access:** Students and parents can preview upcoming topics, helping learners prepare ahead of physical classes.

8. School Rules & Parent Undertaking Compliance

- **Digital Policy Access:** Direct access to school code of conduct, dress codes, and academic requirements.
- **Parent Digital Undertaking:** Online agreement verification tracking signed parent acknowledgments for student compliance.

Technical Architecture & Deployment Highlights

Frontend Framework: React 19 + Vite (Modern Single Page App)	Backend Engine: Node.js + Express REST API
Database Layer: Dual Engine (SQLite / MySQL Connection Pool)	Security & Auth: JWT Session Tokens + Bcrypt Password Hashing
User Interface: Glassmorphism Design + Dark/Light Theme	Deployability: Zero-dependency portable runtime, Docker ready

Value Proposition for Stakeholders

Stakeholder	Key Business & Operational Benefits
School Management & Proprietors	• Complete administrative control over all academic & financial operations. • Accelerated fee collection through Result PIN access control. • Eliminates paper loss and manual report card calculation errors. • Custom branding tailored to Jere Model Academy's identity.
Teachers & Educators	• Reduces report card preparation time by over 80%. • Automatic total and grade computation prevents human calculation mistakes. • Easy digital attendance marking and scheme of work delivery.
Parents & Students	• Instant online result access from any smartphone or computer. • Transparent payment records and official downloadable receipts. • Clear visibility into weekly curriculum and child's behavioral ratings.

Summary Conclusion:

The Jere Model Academy School Management System provides a complete, modern, and reliable digital transformation solution for school administration. It ensures operational accuracy, robust financial control, and high parent satisfaction. Ready for immediate deployment and custom school integration.